

GB

Oxide/metal interface (fixed)

Oxide

Metal

interfacial reactions

Oxygen diffusion
across the oxide layer

J_O : transport along the GB

$J_r^{\text{Cr}_2\text{O}_3}$

$J_r^{\text{Fe}_3\text{O}_4}$

$J_r^{\text{FeCr}_2\text{O}_4}$

$J_r^{\text{SiO}_2}$

$v_{\text{lat}}^{\text{ox}}$ (lattice drift)

